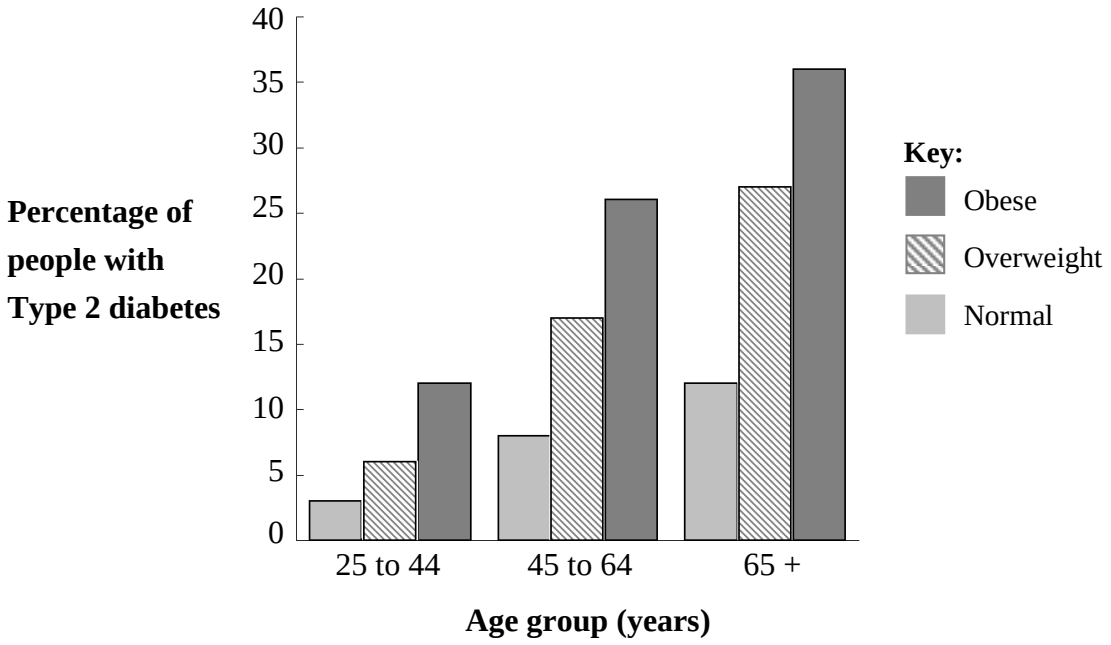
Structured questions

|!|BCELB0240000001|!| ★ **STSE**

The bar chart below shows the percentage of people with Type 2 diabetes in different age groups with different body masses.



(a) Type 2 diabetes is caused by insulin resistance. Explain what is meant by insulin resistance. (2 marks)

(b) Describe the trends shown in the bar chart. (2 marks)

(c) Advise ***two*** lifestyle changes that can help control Type 2 diabetes and explain how they may work. (4 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) In Type 2 diabetes, the pancreas can produce sufficient insulin, (1)
but the target cells (e.g. liver cells and body cells) do not respond to insulin, (1)
resulting in high blood glucose levels.

(b) The incidence of Type 2 diabetes increases with age. (1)
The incidence of Type 2 diabetes is higher in obese / overweight people. (1)

(c) Reduce dietary intake of sugars / fat / high-energy foods. (1)+
Less fat is converted to glucose / less glucose is absorbed into the blood. (1)
Do more exercise. (1)+
More glucose / fat is used in respiration to release energy. / Adequate exercise can help lose weight. (1)

(8 marks)

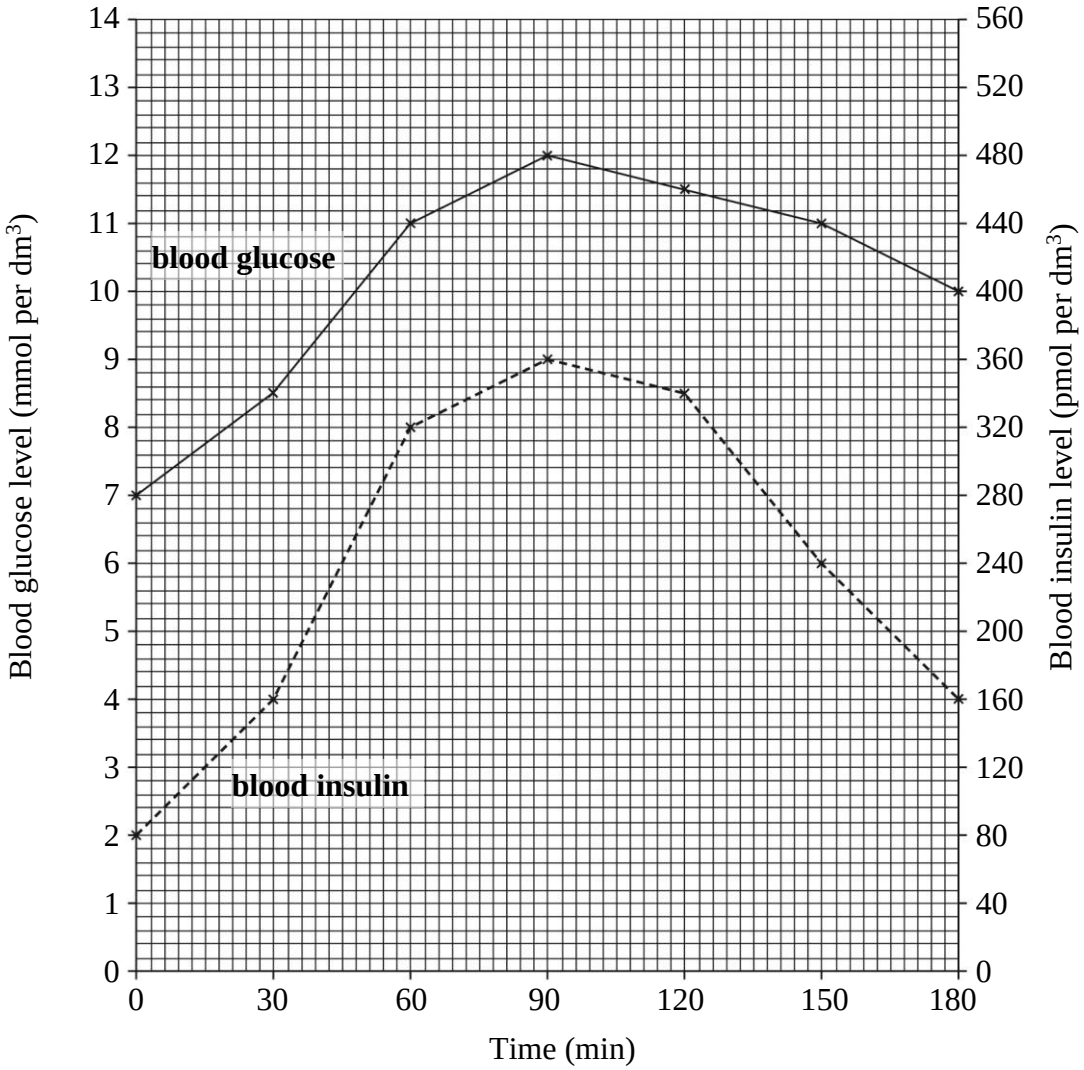
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	2
(c)	Understanding basic concepts	4

© 2020 Aristo Educational Press Ltd.

|!|BCELB0240000002|!| ★ **Cross topic**

The glucose tolerance test is used for diagnosing diabetes. Persons being tested are required not to eat or drink anything except water for 10 hours and then drink a measured dose of glucose solution. Afterwards, blood is taken at intervals for measurement of blood glucose and blood insulin levels. The graph below shows a result of such test.



(a) Calculate the percentage increase in the blood glucose level at the 90th minute compared to the starting level. (2 marks)

_____

_____

(b) The result shows that the tested person has diabetes. Suggest which type of diabetes the person has. Explain your answer. (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) Some diabetic patients can be treated with regular insulin injections.

(i) Is regular insulin injection suitable for treating the type of diabetes you answered in (b)? Explain your answer. (2 marks)

(ii) Insulin for medical use is mostly produced using recombinant DNA technology. Outline the major steps. (5 marks)

##

(a) $(12 - 7) \div 7 \times 100\%$ (1)
$= 71.4\%$ (1)

(b) Type 2 diabetes (1)
The person's blood insulin level rose following the rise in blood glucose level, (1)
but his cells are insensitive to the increased amount of insulin, (1)
so his blood glucose level remained high.

(c) (i) No, (1)
because the person's cells are insensitive to insulin. (1)

(ii) Isolate DNA fragments that contain the human insulin genes from donor cells. (1)
Isolate plasmids to be used as vectors from bacteria. (1)
Cut the DNA fragments and the plasmids using the same restriction enzyme. (1)
Join the cut DNA fragments and the open plasmids together using a DNA ligase. (1)
Introduce the recombinant plasmids into bacteria. The transformed bacteria express the human insulin gene and synthesize human insulin. (1)

© 2020 Aristo Educational Press Ltd.

(12 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	2
	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	5

|!|BCELB0240000003|!| ★ **Cross topic**

The table below lists the top 10 causes of death in low-income countries and high-income countries in 2016.

Low-income countries		**High-income countries**	
Cause	**% of total deaths**	**Cause**	**% of total deaths**
Lower respiratory infections	9.3	Coronary heart disease	16.8
Diarrhoeal diseases	7.2	Stroke	7.2
Coronary heart disease	6.5	Alzheimer disease and other dementias	7.0
HIV/AIDS	5.5	Trachea, bronchus, lung cancers	5.6
Stroke	5.2	Chronic obstructive pulmonary disease	5.4
Malaria	4.6	Lower respiratory infections	4.3
Tuberculosis	4.2	Colon and rectum cancers	3.2
Preterm birth complications	4.0	Diabetes mellitus	2.7
Birth asphyxia (窒息) and birth trauma	3.8	Kidney diseases	2.2
Road injury	3.6	Breast cancer	1.8

(Data source: World Health Organization)

(a) Pneumonia and bronchitis are examples of lower respiratory infections. Suggest one way by which these infections may spread. (1 mark)

(b) How does coronary heart disease develop in the human body? (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) Give ***one*** difference between the top 10 causes of death in low-income countries and high-income countries. (2 marks)

(d) Explain for the difference you have answered in (c) with reference to the general conditions in low-income countries. (3 marks)

(e) Explain for the difference you have answered in (c) with reference to the lifestyles in high-income countries. (3 marks)

##

© 2020 Aristo Educational Press Ltd.

(a) These infections may be transmitted through droplets or air. (1)

(b) Excess cholesterol and other fatty substances in blood deposit on the inner walls of coronary arteries. (1)

The deposits (plaque) narrow the lumen of the coronary arteries and reduce the blood flow to cardiac muscle. (1)

This reduces the oxygen and nutrient supply to the cardiac muscle. (1)

(c) Five out of the top 10 causes of death in low-income countries are infectious diseases, (1) +

while eight out of the top 10 causes of death in high-income countries are non-infectious diseases. (1)

(Or other reasonable answers)

(d) Any ***three*** of the following: (3)

- Overcrowded living conditions favour the spread of airborne diseases (e.g. measles, tuberculosis, etc.).
- Drinking water is likely to be contaminated due to poor sewage treatment. This favours the spread of water-borne diseases (e.g. diseases that lead to diarrhoea).
- Mosquitoes thrive in some low-income countries in the tropical areas and mosquitoes can spread malaria.
- Due to poverty, people in low-income countries cannot afford vaccines to prevent the spread of infectious diseases / medical costs to treat infectious diseases.

(Or other reasonable answers)

(d) Any ***three*** of the following: (3)

- Consumption of fatty foods increases the risk of plaque formation, coronary heart disease and stroke.
- People are able to afford cigarettes and alcohol. Heavy smoking increases the risk of lung cancer.Alcohol abuse increases the risk of liver cancer.
- High-fat and low-fibre diets increase the risk of colorectal cancer.
- Unhealthy diet and lack of exercise increase the risks of cardiovascular diseases, various types of cancers and type 2 diabetes.

(Or other reasonable answers)

(12 marks)

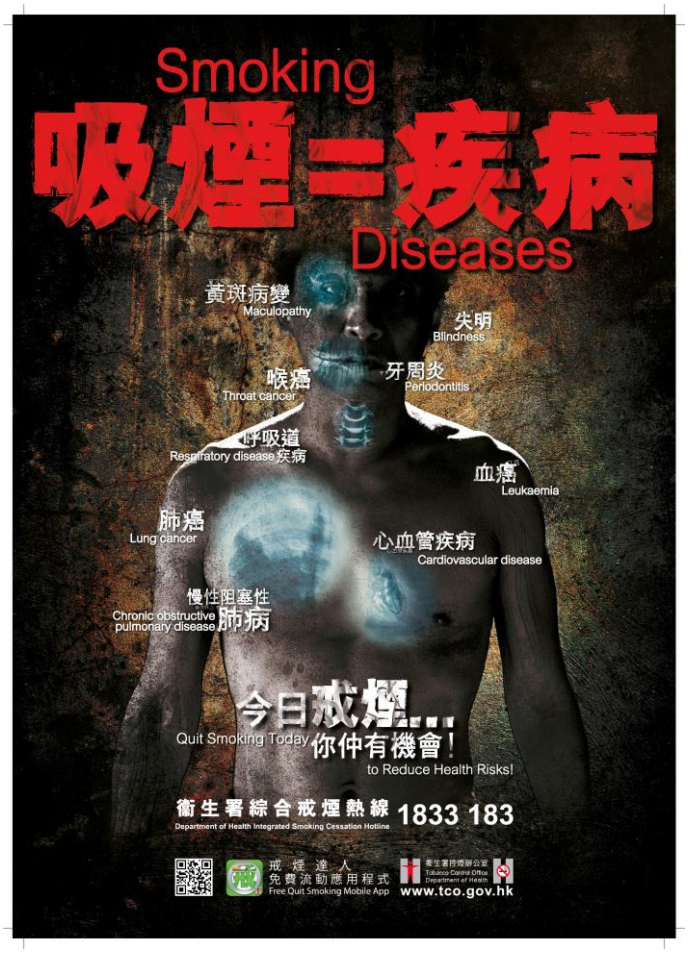
##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Understanding basic concepts	3
(c)	Analysing data	2
(d)	Applying concepts	3
(e)	Applying concepts	3

© 2020 Aristo Educational Press Ltd.

|!|BCELB0240000004|!| ★ **Cross topic**

The poster below is made by the Tobacco and Alcohol Control Office of the Department of Health to encourage smokers to quit smoking.



(a) Chemicals in cigarette smoke damage the walls of air sacs. Explain how this affects the efficiency of gas exchange in smokers' body. (2 marks)

(b) Studies showed that smoking causes lung cancer.

(i) State a substance in cigarette smoke that is cancer-causing. (1 mark)

(ii) Describe how carcinogens in cigarette smoke may lead to lung cancer. (3 marks)

© 2020 Aristo Educational Press Ltd.

(c) Explain how smoking may increase the risk of cardiovascular diseases. (4 marks)

##

(a) Air sacs are the site of gas exchange. A reduction in the surface area of air sacs lowers the efficiency of gas exchange. (1) (1)

(b) (i) tar (1)

(ii) Carcinogens in cigarette smoke induce mutations in the cells lining the respiratory tract. (1)

When the genes that regulate cell division are mutated, lung cells may divide uncontrollably and excessively. (1)

When the lung cells undergo further mutations, they become cancer cells. (1)

(c) Nicotine in cigarette smoke increases blood viscosity and the chance of forming blood clots. (1)

Blood clots may block any narrowed blood vessels. (1)

Nicotine causes blood vessels to constrict and raises the heart rate. (1)

This forces the heart to work harder and increases the risk of heart attack. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3
(c)	Understanding basic concepts	4

© 2020 Aristo Educational Press Ltd.

|!|BCELA0240000005|!| ★★

Familial hypercholesterolemia (FH) is a genetic disease that is characterized by much higher blood cholesterol levels than normal people because of poor metabolism of cholesterol by the liver.

(a) Explain why patients with FH are likely to develop coronary heart disease at a much younger age than the general population. (3 marks)

(b) Outline how coronary heart disease can be treated with balloon angioplasty. (3 marks)

(c) Give ***two*** pieces of dietary advice to people with a high blood cholesterol level to help lower the risk of coronary heart disease. (2 marks)

##

(a) The chance of cholesterol deposition on the inner walls of coronary arteries is higher in patients with FH due to their high blood cholesterol levels. (1)
The rate of plaque building up is faster in patients with FH. (1)
Therefore, the affected coronary arteries will become severely narrowed or completely blocked within a shorter period of time. (1)

(b) In balloon angioplasty, a small tube with a tiny balloon is inserted into the narrowed region of the artery. (1)
The balloon is inflated to press against and flatten the plaque. (1)
This widens the lumen of the artery to restore the blood flow. (1)

© 2020 Aristo Educational Press Ltd.

(c) Any ***two*** of the following: (2)

- Reduce the intake of cholesterol and fat / saturated fats / animal fat.
- Obtain essential fatty acids from plant sources and fish.
- Increase the consumption of foods that help lower the blood cholesterol level, such as oatmeal and other high-fibre foods.

(Or other reasonable answers)

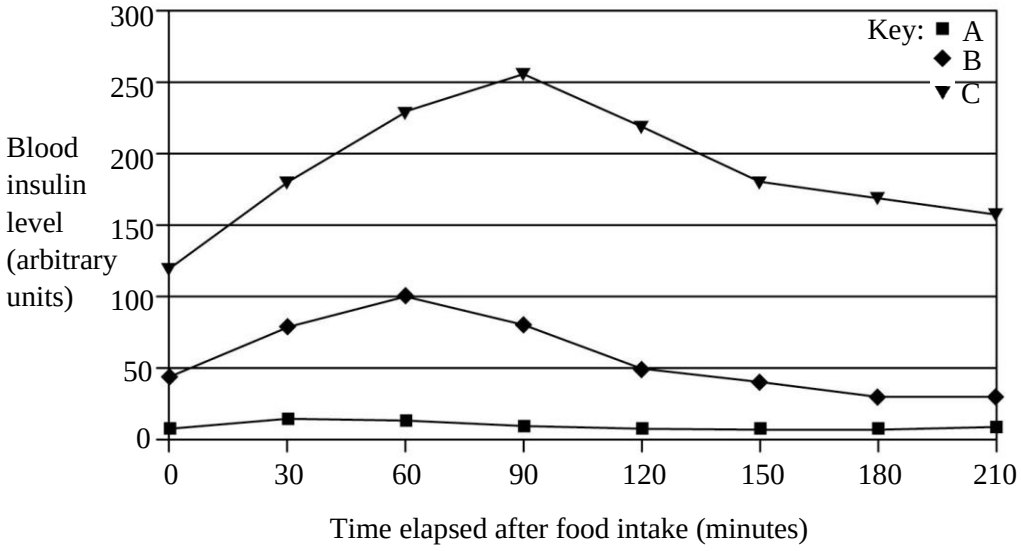
(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Understanding basic concepts	3
(c)	Applying concepts	2

|!|BCELB0240000006|!| ★ **Cross topic**

The graph below shows the changes in the blood insulin levels of persons A, B and C after food intake. It is known that person B is healthy, while persons A and C are diabetic.



(a) With reference to the graph, deduce which type of diabetes do persons A and C have, respectively. (6 marks)

© 2020 Aristo Educational Press Ltd.

(b) Suggest ***one*** medical treatment for persons A and C, respectively. (2 marks)

(c) Referring to the homeostatic control of blood glucose in the body, describe how a normal blood glucose level is restored after food intake in person B. (4 marks)

##

(a) For person A, his blood insulin level is always lower than normal even after food intake. (1)
This shows that little or no insulin is produced by person A's pancreas. (1)
He is likely to suffer from Type 1 diabetes (insulin-dependent diabetes). (1)
For person C, his blood insulin level is elevated after food intake, but his blood glucose level is still abnormally high. (1)
This shows that his body (liver) shows no response to insulin. (1)
He is likely to suffer from Type 2 diabetes (non-insulin-dependent diabetes). (1)

(b) Person A: regular injection of insulin (1)
Person C: restricting the dietary carbohydrate intake / weight management / regular exercise (1)

(c) A rise in the blood glucose level after food intake is detected by the pancreas. (1)
The pancreas secretes more insulin into the blood. (1)
Insulin stimulates the liver cells to convert excess glucose to glycogen / the uptake of glucose by body cells and subsequent oxidation of glucose by respiration. (1)
Hence, the blood glucose level falls back to normal. (1)

(12 marks)

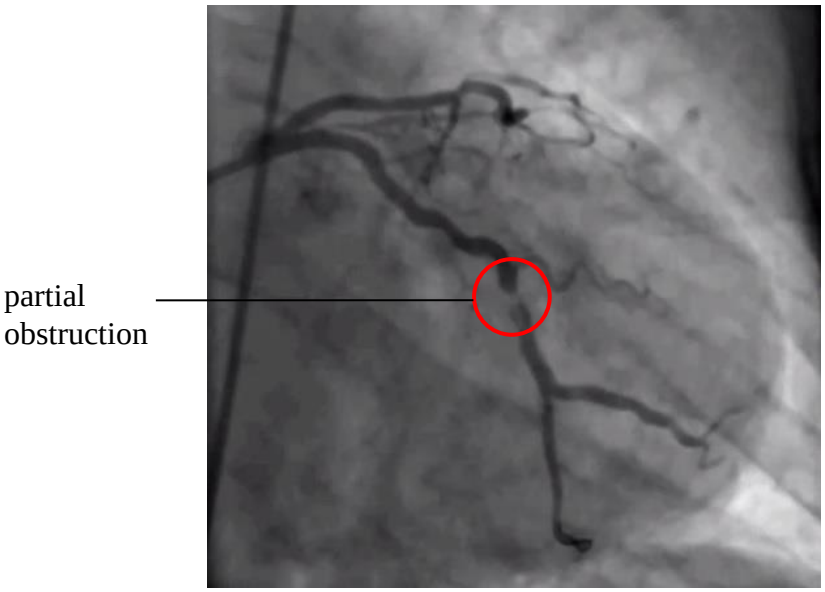
© 2020 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	4
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	4

|!|BCELB0240000007|!| ★ **STSE**

David occasionally has angina (chest pain), so he went to see a doctor. The doctor used X-rays to check the condition of David's coronary arteries. The image below shows the results.



Source: https://commons.wikimedia.org/wiki/File:Coronary_angiography_of_a_STEMI_patient,_showing_partial_occlusion_of_left_circumflex_coronary_artery.jpg

Credit: Todt T, Maret E, Alfredsson J, Janzon M, Engvall J, Swahn E

(a) One of David's coronary arteries shows partial obstruction. Using your biological knowledge, describe how a coronary artery may become obstructed. (3 marks)

© 2020 Aristo Educational Press Ltd.

(b) The doctor warns David that he has a high risk of heart attack during vigorous excerise. Explain for the high risk of heart attack during vigorous excerise based on the condition of David's coronary arteries. (3 marks)

(c) The doctor advises David to quit smoking. Explain why quitting smoking may help prevent David's condition from worsening. (4 marks)

##

(a) When cholesterol and other fatty substances in blood deposit on the inner wall of a coronary artery, plaque is formed. (1)
Over time, the plaque builds up / becomes larger in size (1)
and blocks the lumen of the coronary artery. (1)

(b) During vigorous exercise, the heart needs to pump faster and harder to increase the rate of blood circulation. (1)
Narrowing of the coronary artery reduces blood flow to the cardiac muscle. (1)
The cardiac muscle may die when it cannot obtain enough oxygen and nutrient from the blood. (1)

(c) Nicotine in cigarette smoke increases blood viscosity and the chance of forming blood clots. (1)
By quitting smoking, David's chance of plaque formation due to nicotine can be reduced. (1)
Carbon monoxide in cigarette smoke reduces the oxygen-carrying capacity of blood. (1)
By quitting smoking, the amount of carbon monoxide inhaled is reduced and this helps improve blood oxygenation. (1)

© 2020 Aristo Educational Press Ltd.

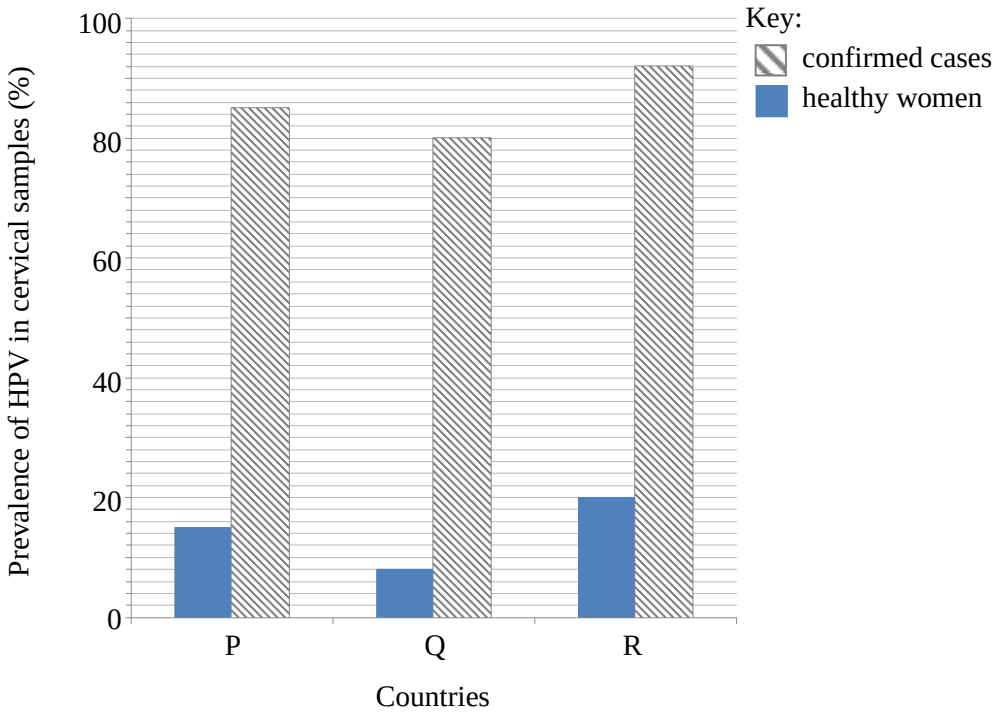
(10 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	3
(c)	Applying concepts	4

|!|BCELA0240000008|!| ★★

An investigation was carried out to study the relation between the Human papilloma virus (HPV) and the incidence of cervical cancer. Cervical samples were collected from 300 confirmed cases of cervical cancer and 300 healthy women from three countries (P, Q and R). Cells from the samples were tested for the presence of HPV. The prevalence of HPV in the samples is shown in the bar chart below.



(a) Describe evidence from the graph that supports a link between HPV and cervical cancer. (2 marks)

© 2020 Aristo Educational Press Ltd.

(b) Suggest how HPV may cause cervical tissues to become cancerous. (3 marks)

(c) Not all women infected with HPV develop cervical cancer. Give ***two*** other risk factors that increase the chance of developing cervical cancer. (2 marks)

(d) To prevent cervical cancer, the Centre for Health Protection of the Department of Health provides HPV vaccines under the Hong Kong Childhood Immunization Programme to primary school girls starting from the 2019/20 school year.
Give ***one*** benefit and ***one*** ethical concern associated with the provision of HPV vaccines to primary school girls. (2 marks)

##

(a) The prevalence of HPV in the samples from confirmed cases is much higher than that from healthy women. (1)
The higher prevalence of HPV in the samples from confirmed cases is observed in all of the three countries. (1)

(b) The nucleic acids of HPV will enter cervical cells (1)
and may cause mutation that affects the genes that regulate cell division. (1)
The mutated cells will divide excessively and uncontrollably, forming a malignant tumour. (1)

(c) Any ***two*** of the following: (2)
- Smoking
- Family history
- Weakened immune system due to ageing / HIV infection
- Long term use of oral contraceptive pills for more than 5 years

(Or other reasonable answers)

(d) Any ***one*** of the following benefits: (1)
- Women will have immunity against HPV before they become sexually active.

© 2020 Aristo Educational Press Ltd.

- A higher proportion of women can become immunized against HIV through the immunization programme, hence the transmission of HPV infection in the community can be reduced.

(Or other reasonable answers)

Any ***one*** of the following ethical concerns: (1)

- The HPV vaccines may have unknown side effects / long-term adverse effects on health.
- Early vaccination against HPV may encourage sexual intercourse at an earlier age.
- Early vaccination against HPV may encourage having multiple sexual partners.
- The vaccinated women may not have regular screening for cervical cancer.

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	3
(c)	Applying concepts	2
(d)	Critical thinking	2

© 2020 Aristo Educational Press Ltd.